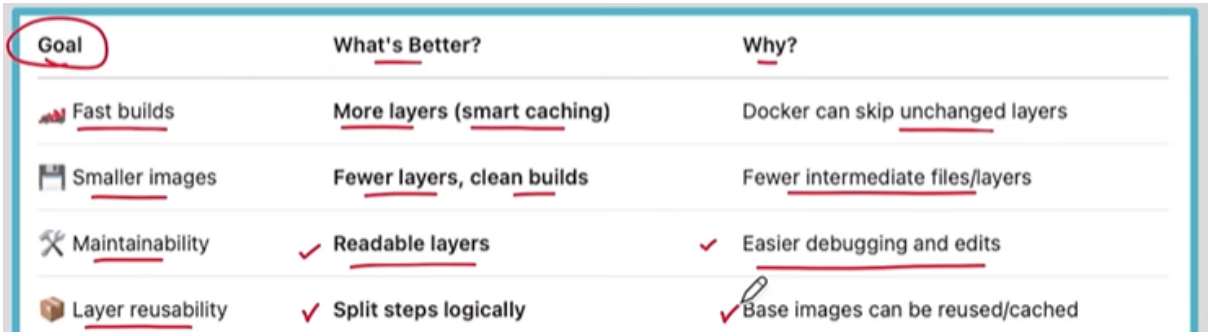
Docker Image Layers - Best Practice Tips

1. Group COPY and RUN wisely to improve layer caching
2. Put less frequently changes lines early
3. Avoid unnecessary RUN commands to reduce layer count and images size



Docker Containers - Runtime
Container runtime is the container execution phase
The image is used to create the container and execute the CMD or ENTRYPOINT command

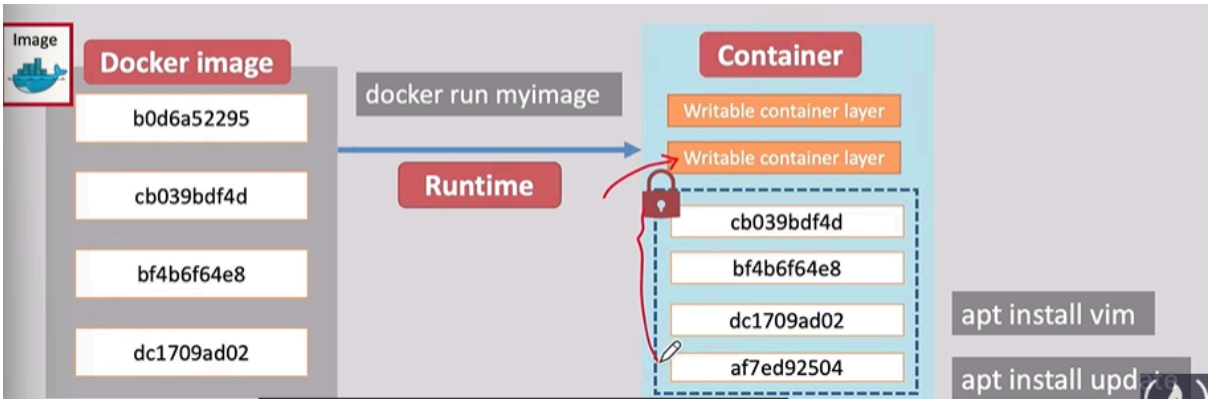


Image tag
Docker run postgres:10.10->image tag

Docker Registry-pushing Images

Docker Hub has a specific naming convention for docker images that helps with organisation , versioning and discovery.

[registry/][Username_or_tag/]repository[:tag]

Registry : the domain of the container registry
Username : the name of the user
Repo : the name of the image or project
Tag(optional) : the version

Image Rename - docker tag Command

The docker tag command is used to add new name to an existing Docker image

Docker tag python:30 my-python-app
GIve the image python.30 a custom name my-python-app:latest

Docker Registry- Docker tag command

Image must be correctly tagged with the intended registry & image name before they pushed

Docker tag myapp default/devops/myapp:1.0.0
This tag image myapp with the name im preparation to be pushed to the private registry under account devops under repo/image myapp with the tag 1.0.0

Docker login command

docker login Registry_Host

DockerHub registry
Docker tag LOCAL_IMAGE:tag USERNAME/REPO:TAG

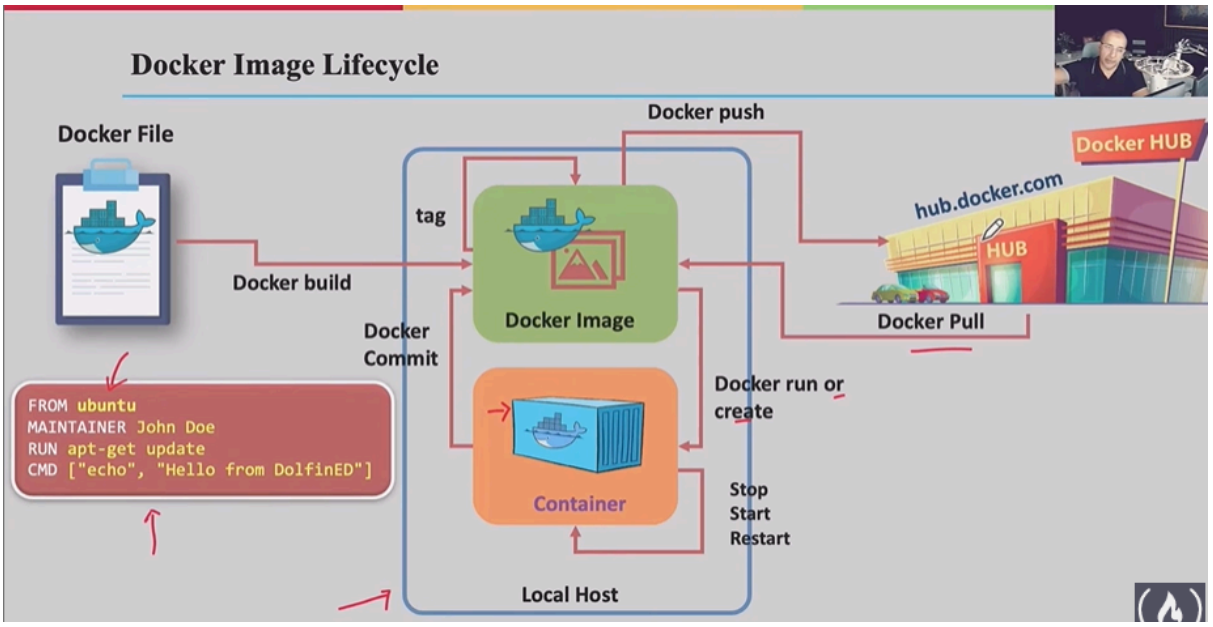
Docker tag myimage myusername/myapp:latest
Myimage is tagged in preparation to docker hub registry , under account myusername under the repo myapp with tag latest

Docker Registry - docker push Command

Docker push uploads a locally tagged image to docker registry

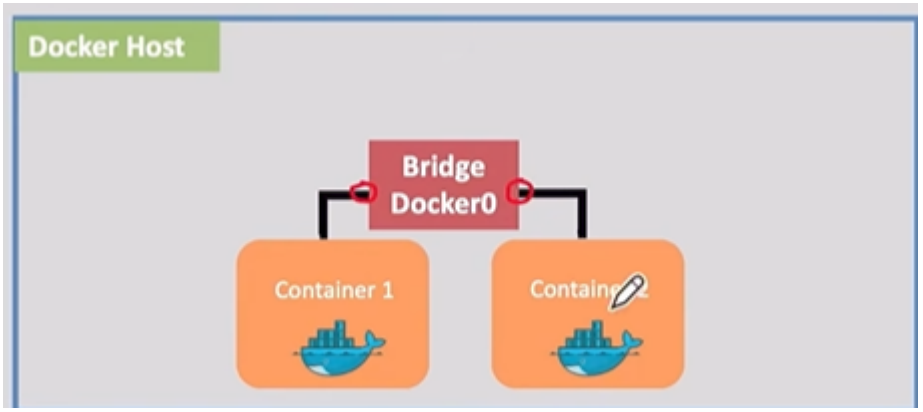
Docker push [REGISTRY_HOST]/USERNAME/REPO:TAG

Docker push myapp myusername/myapp:latest



Docker Bridge Networks

In docker a bridge network uses a software bridge to allow containers to communicate inside the docker host.
By default when you start Docker , a bridge (docker0) is created automatically.

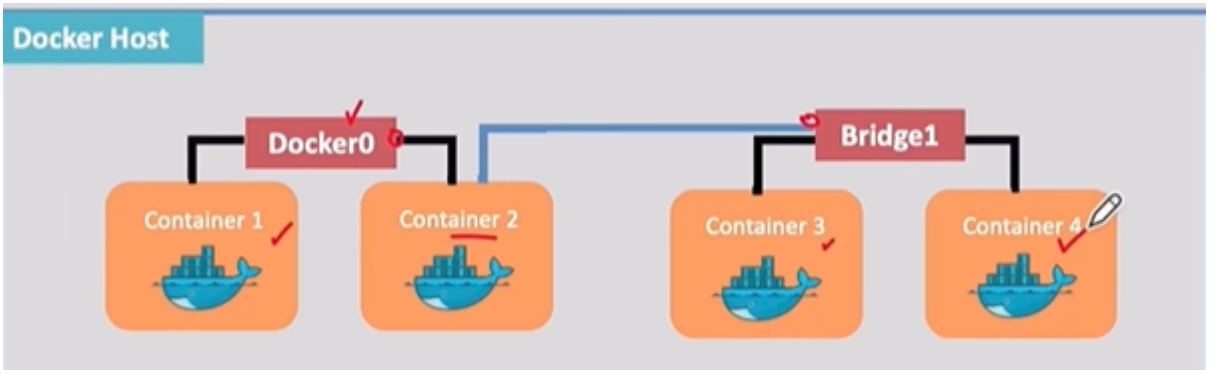


Docker Network Modes
Bridge(default)
Host
None
Overlay
Ipvlan
Macvlan
Userdefined

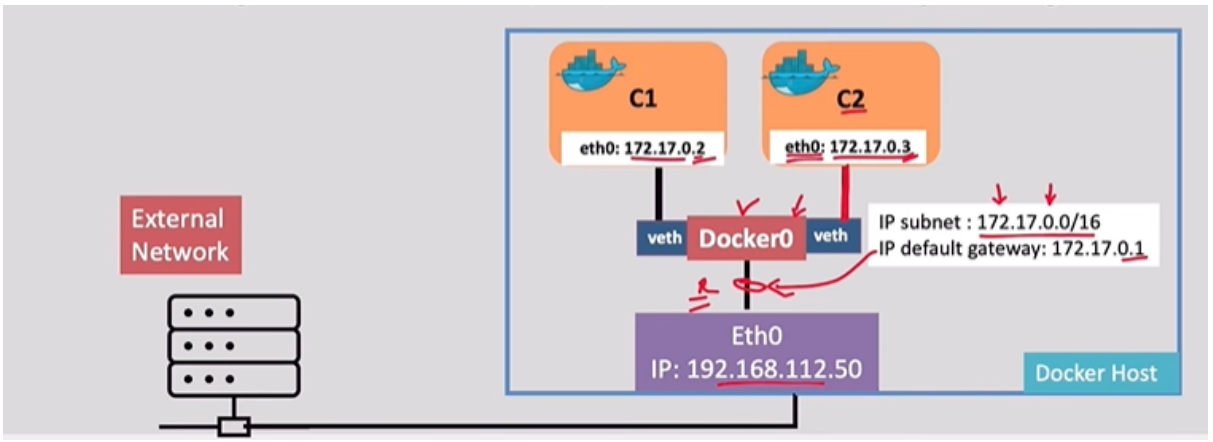
Bridge Driver/Mode - container and Docker Bridge Networks

>Containers connected to the same bridge network can communicate

A bridge network isolates its containers from other containers on other bridges
Bridge network apply within the same docker host
A container can connect to multiple bridge networks at the same time

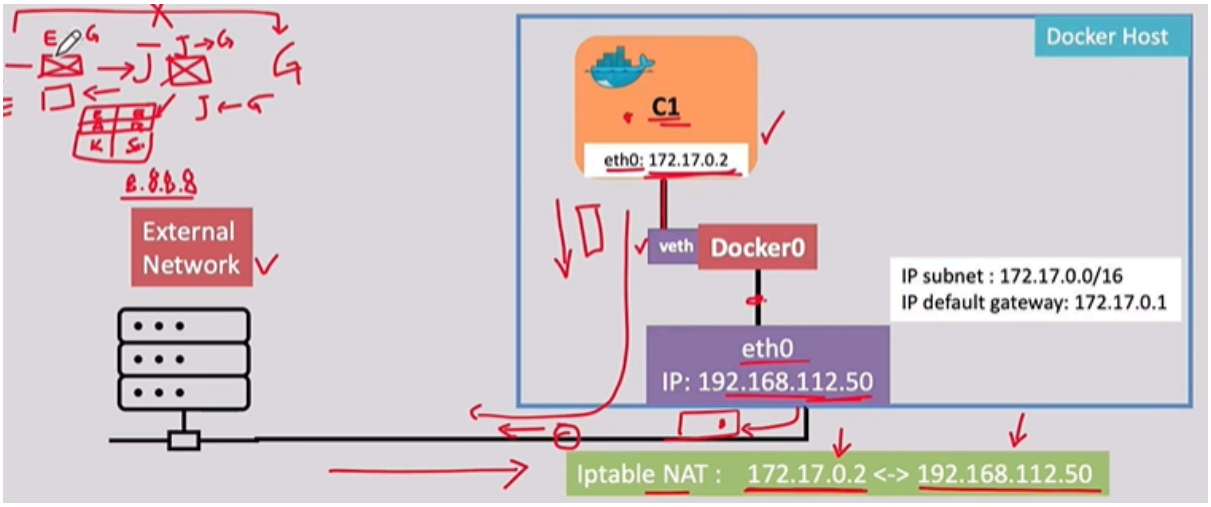


Docker Default Bridge(docker0)
The default docker bridge network (docker0) IP subnet is 172.17.0.0/16
172.17.0.1 is the IP default gateway between bridge Docker0 and th Docker host
Unless otherwise specified all newly created containers attach th Docker0
Each container created in bridge mode has an eth0 with an assigned IP address
A new bridge virtual interface (veth) is created with every running container

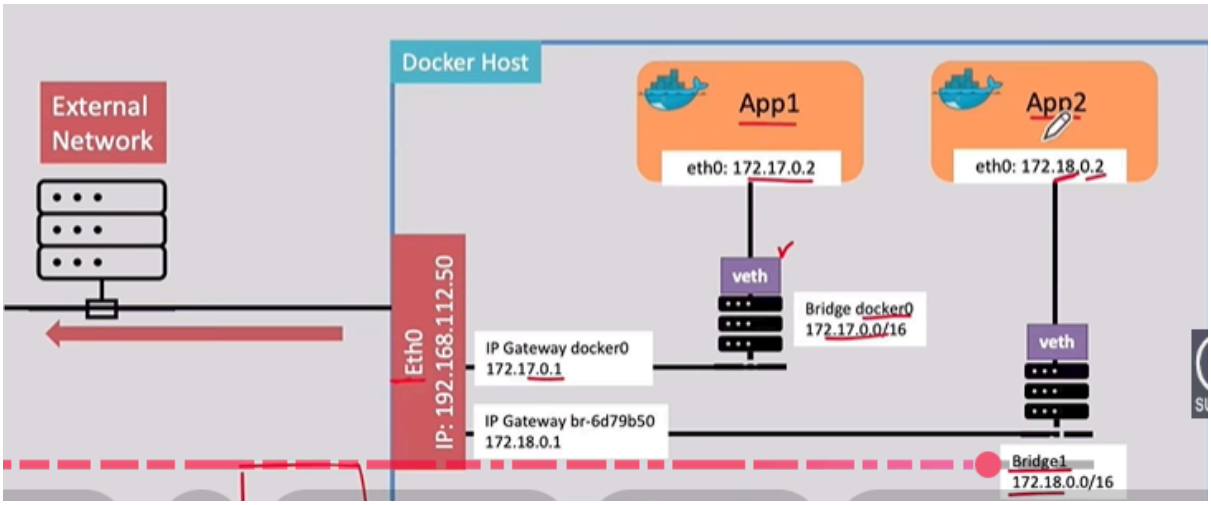


Bridge Mode - Containers outbound Connectivity To External Network

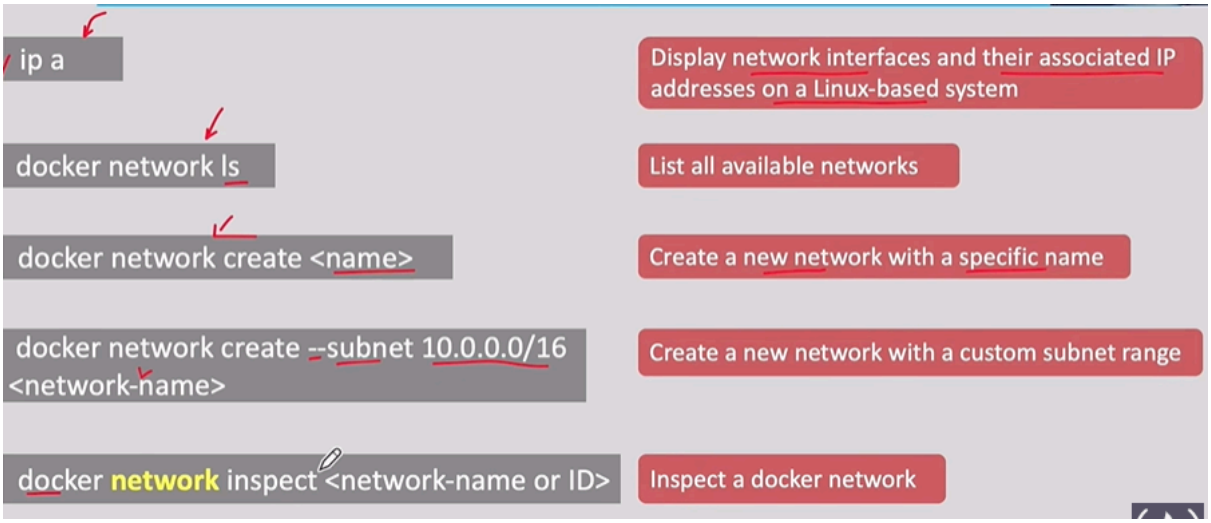
In bridge network mode , when a cont. Communicates outbound with external networks the docker host performs Network Address Translation(NAT)
The Docker host maintains NAT entries in IPtables



Bridge Mode - Multiple Bridge Networks
Multiple bridge networks can co-exist inside a docker host
We can create custom bridge networks as needed (aside from the default bridge)
A bridge network isolates its containers from other containers on other bridges



Docker Network Commands



Docker Network Command (cont.)

